

Project Design Phase-II Technology Stack (Architecture & Stack)

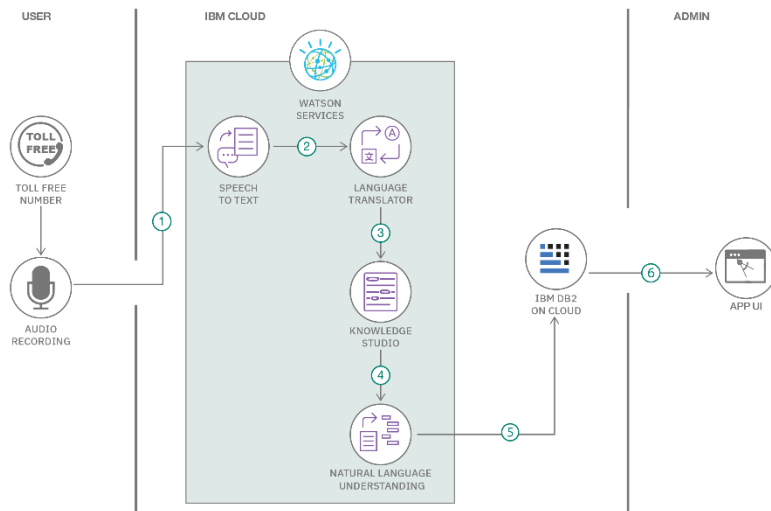
Date	10 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Users interact with the app through a web browser.	React.js, Tailwind CSS
2.	Application Logic-1	Handles user authentication - register and login	Node.js, Express.js, JWT, bcryptjs
3.	Application Logic-2	Handles product catalog, cart and order management	Node.js, Express.js, Mongoose
4.	Database	Stores users, products, carts and orders	MongoDB
5.	File Storage	Product images loaded from Unsplash URLs	External image URLs
6.	External API-1	Product images fetched from Unsplash	Unsplash Image API
7.	Infrastructure (Server / Cloud)	Running locally on developer machine	localhost:3000

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	All frameworks used are open source	React.js, Node.js, Express.js, MongoDB, Mongoose, Tailwind CSS
2.	Security Implementations	Passwords are hashed and JWT tokens used for auth	bcryptjs for hashing, JWT for authentication.
3.	Scalable Architecture	Frontend and backend are decoupled	React frontend and Express backend run independently
4.	Availability	App runs locally, can be deployed to cloud	Vercel for frontend, Render for backend
5.	Performance	Lightweight API calls using Axios	Axios for HTTP requests, MongoDB for fast queries

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>